

Task

1. Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

Linux (BIND) Install

BIND: `sudo apt`

`update sudo apt`

`install bind9`

Create a new DNS zone:

- Zone Name: musicmania.in

Edit the BIND configuration file: `sudo`

`nano /etc/bind/named.conf.local`

Add: zone

```
"musicmania.in" {  
    type master;    file  
    "/etc/bind/db.musicmania.in"; };
```

Windows (Simple DNS Plus)

1. Install Simple DNS Plus.
2. Open the application.
3. Click New Zone.
4. Enter the zone name:

musicmania.in

5. Save the configuration.

2. Add an A record in your DNS server configuration to map **www.musicmania.in** to the IP address **192.168.1.50**, then use the **nslookup** or **dig** command to verify the mapping works.

DNS Record

Record Type	Host Name	IP Address
A	www.musicmania.in	192.168.1.50

Verify Windows nslookup

www.musicmania.in Linux

dig www.musicmania.in

Expected Output

Name: www.musicmania.in

Address: 192.168.1.50

3. Simulate a typo by adding a CNAME record for **play.musicmania.in** that points to **www.musicmania.in**, then test if accessing **play.musicmania.in** resolves to the correct IP address.

DNS Record

Record Type	Alias	Points To
CNAME	play.musicmania.in	www.musicmania.in

Verify nslookup **play.musicmania.in**

Expected Output **play.musicmania.in**

canonical name = **www.musicmania.in**

Address: 192.168.1.50

This shows that play.musicmania.in successfully resolves to 192.168.1.50 through the CNAME record.

4. Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.
Hint: Include at least three different troubleshooting commands or checks in your documentation.

Step 1: Check DNS Configuration

Verify that the zone file contains the correct records.

Linux `sudo nano /etc/bind/db.musicmania.in`

Windows

- Open Simple DNS Plus.
- Check that the A and CNAME records are correctly configured.

Step 2: Restart the DNS Service

Linux `sudo systemctl restart`

`bind9`

Windows

- Restart the Simple DNS Plus service.

Step 3: Verify DNS Resolution Windows

`nslookup musicmania.in`

Linux

`dig musicmania.in`

Step 4: Check Service Status (Linux)

`sudo systemctl status bind9` **Step 5:**

Test Network Connectivity ping

192.168.1.50

Explanation

If musicmania.in is not resolving:

1. Check the DNS zone configuration for errors.
2. Restart the DNS service to apply changes.
3. Test using nslookup or dig to verify the records.
4. Check the DNS service status to ensure it is running.
5. Verify that the destination server (192.168.1.50) is reachable using ping